Subject: ACS 547, Module 4 Assignment - Sound Sources

Date: April 18, 2025 (Submitted)

Problem 1 - Simulation of Sources from Monopoles

The Matlab code for this problem is listed in Appendix 1.

Problem 1a

Figure 1 shows the spherical spreading from a monopole. The frequency is 1 kHz. The sound pressure level has been calculated at two distances: 1 m and 4 m.

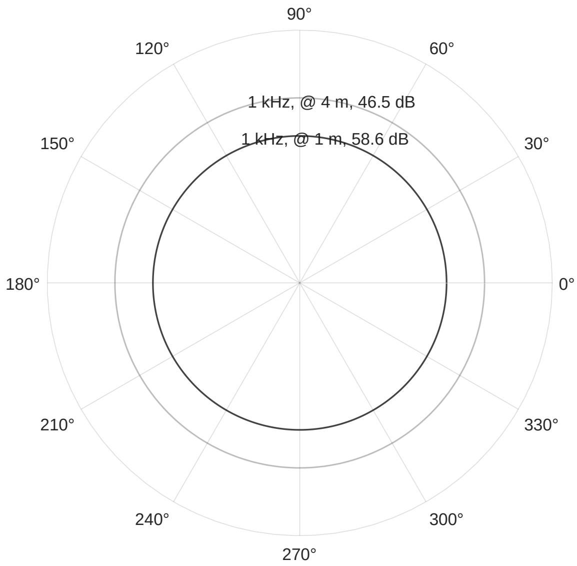

Figure 1: Monopole spherical spreading for 1 kHz. The radius of the polar plot is the sound pressure level.

Figure 2 shows the sound pressure level as a function of distance for a 1 kHz monopole source.

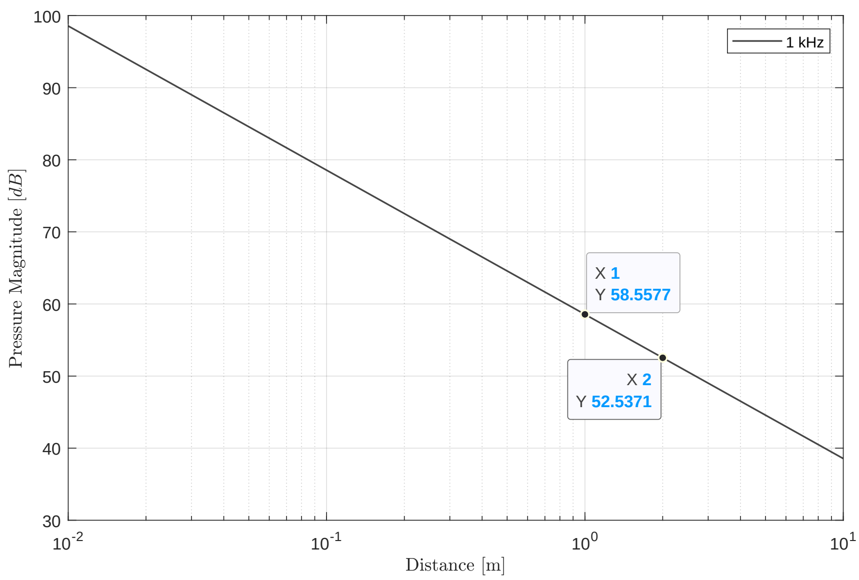

Figure 2: Monopole pressure drop versus distance for 1 kHz.

The pressure drop is proportional to $\frac{1}{r}$; doubling the distance yields -6 dB of level change (see data points).

Problem 1b

Figure 3 illustrates the directivity patterns for a monopole, dipole, lateral quadrupole, and a linear quadrupole for 1 kHz. The receivers are placed at a 1 m distance around the sources.

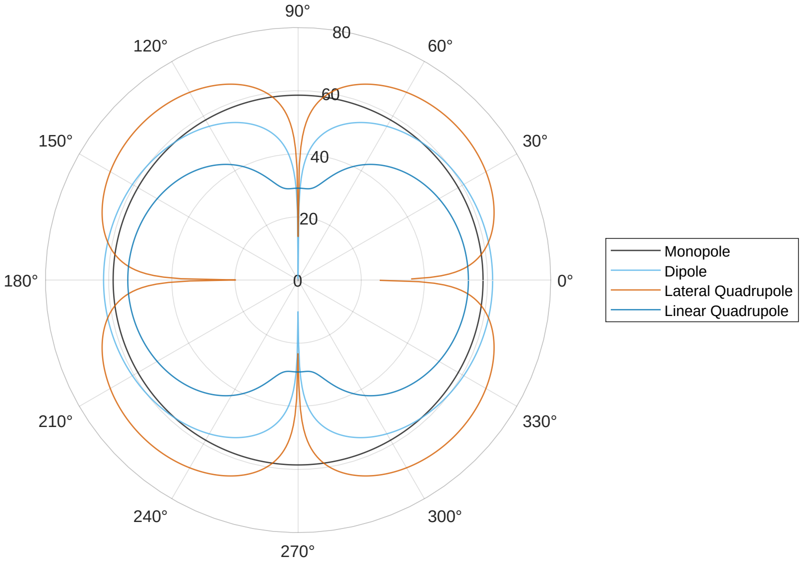

Figure 3: Directivity patterns for a 1 kHz monopole, dipole, lateral quadrupole, and linear quadrupole.

Problem 1c

Figure 4 shows the directivity patterns for a line source with 9 sources separated by 0.043 m centered at the origin along the x-axis for 100 Hz, 2 kHz, and 5 kHz. The receivers are placed at a 1 m distance around the sources.

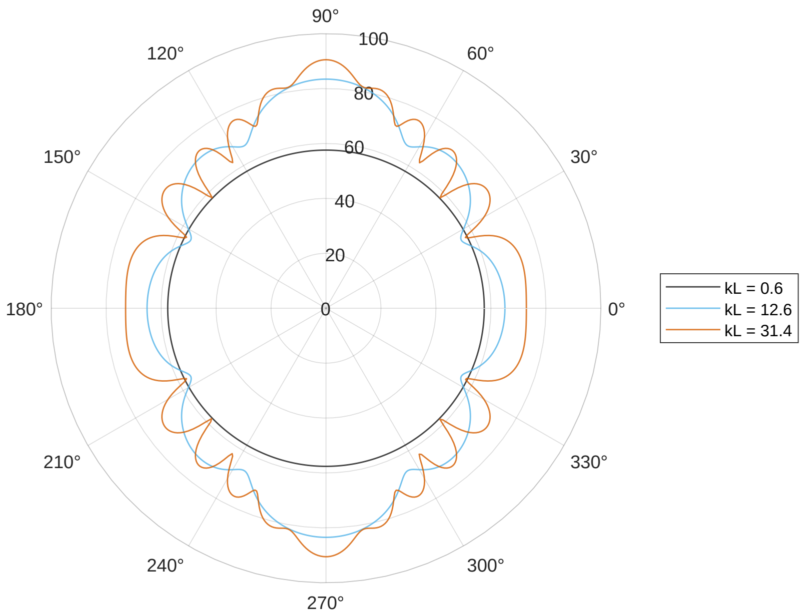

Figure 4: Line source directivity patterns for frequencies of 100 Hz, 2 kHz, and 5 kHz with small source separation.

Figure 5 shows the directivity patterns for a line source with 9 sources separated by 0.343 m centered at the origin along the x-axis for 100 Hz, 2 kHz, and 5 kHz. The receivers are placed at a 1 m distance around the sources.

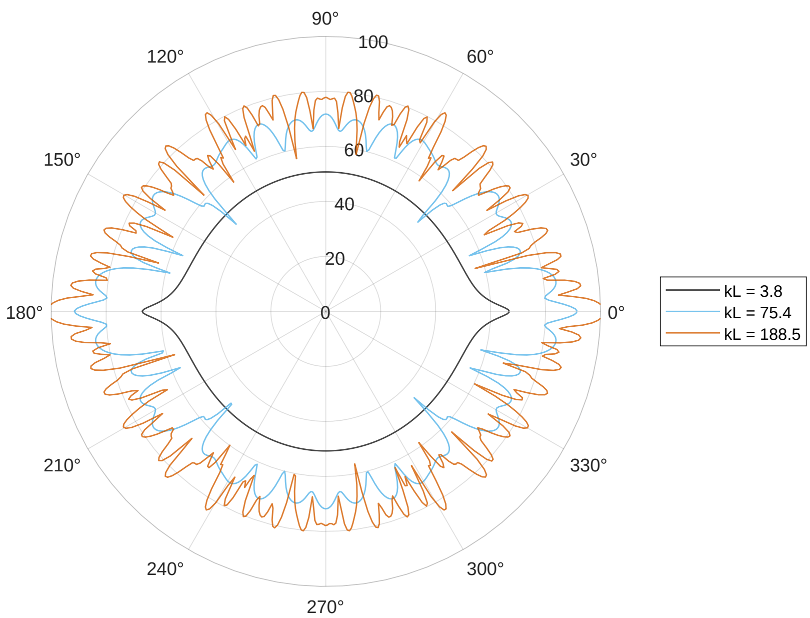

Figure 5: Line source directivity patterns for frequencies of 100 Hz, 2 kHz, and 5 kHz with large source separation.

Problem 1d

Figure 6 shows the monopole source locations in the xy-plane.

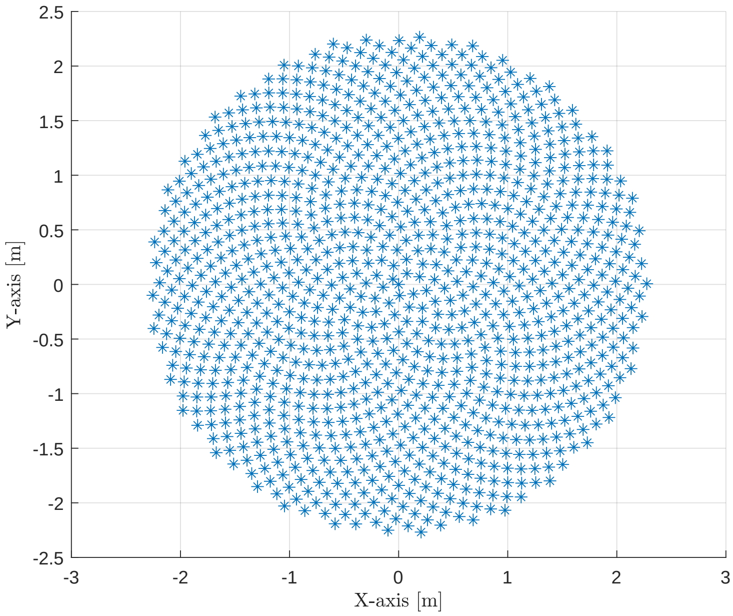

Figure 6: Monopole source locations in the xy-plane.

Figure 7 shows the monopole source locations in the xy-plane.

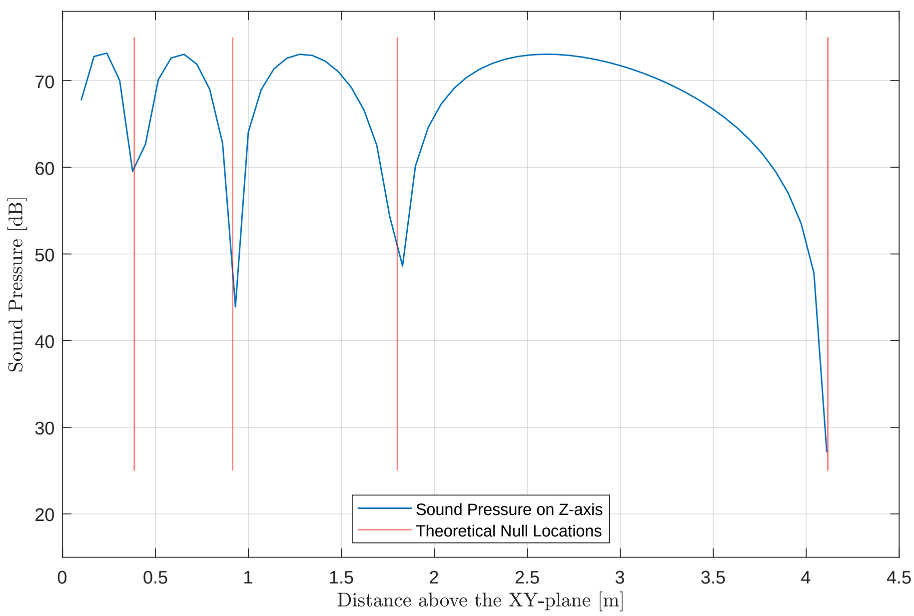

Figure 7: Monopole source locations in the xy-plane.

Problem 2 - Active Control of Enclosure Opening

The Matlab code for this problem is listed in Appendix 2.

Problem 2a

Figure 1 shows the sound power level reduction versus octave band frequency for the linear quadrupole system.

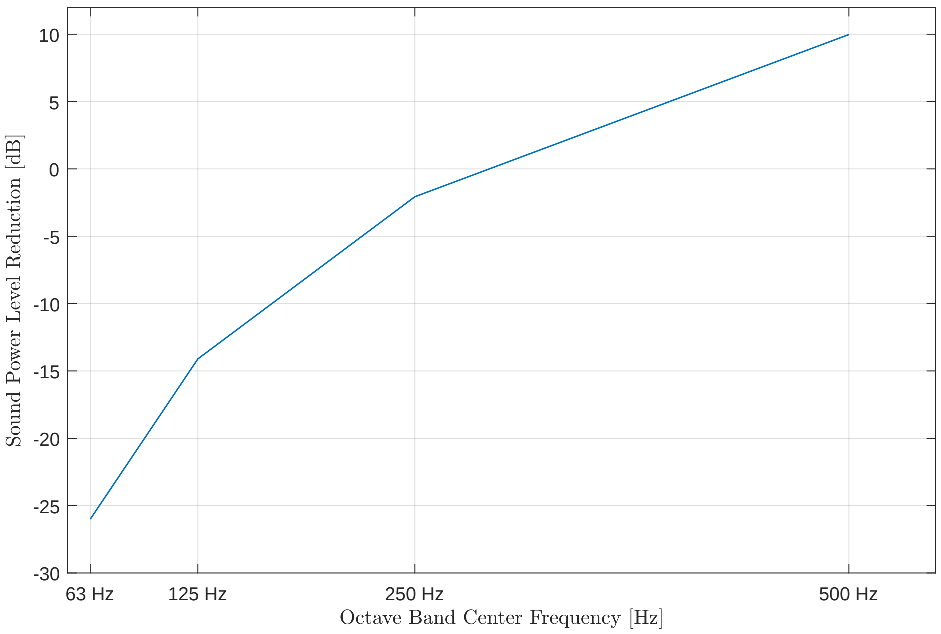

Figure 1: Sound power level reduction as a function of octave band center frequency.

add intuitive description of what is happening

Problem 2b

Figure 2 shows the sound power level reduction versus octave band frequency for the linear quadrupole system.

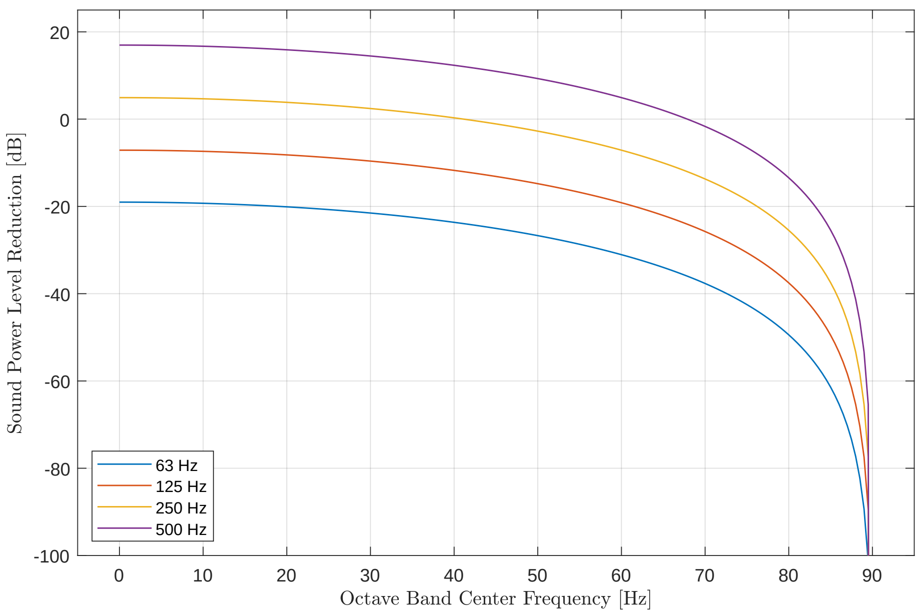

Figure 2: Sound power level reduction as a function of octave band center frequency.

add intuitive description of what is happening

Problem 3 - Washing Machine Dynamic Vibration Absorber Design

Part b - The three speakers have inde

The Matlab code for this problem is listed in Appendix 3.

Part 3a

Figure ?? illustrates the dynamic vibration absorber (DVA) system.

Table 1 lists the stiffness and damping coefficients for the system.

M1	15	kg
K1	0	$\frac{\text{N}}{\text{m}}$
C1	0	$\frac{\text{kg}}{\text{s}}$

M2	110	kg
K2	9,810	$\frac{\text{N}}{\text{m}}$
C2	1,962	$\frac{\text{kg}}{\text{s}}$[1]

M3	0	kg
K3	1e5	$\frac{\text{N}}{\text{m}}$
C3	1e4	$\frac{\text{kg}}{\text{s}}$[2]

Table 1: Stiffness and damping coefficients for the DVA system.

Damping was implemented by adding with an imaginary component to the respective spring stiffness[3].

[1] 0.2 fraction of respective spring stiffness.
[2] 0.1 fraction of respective spring stiffness.
[3] Not equivalent to using a damping vector in the argument to the given Matlab nDOF function script-file.

Part 3b

Figure ?? shows the admittance response of the original system and DVA system based on the equations presented in the tutorial lecture on DVA system analysis.

The washing machine admittance at 300 RPM is reduced from the blue curve resonance peak to a point between the two resonances on the red curve.

Part 3c - Admittance

Figure ?? shows the admittance response of the DVA system using the nDOF function script-file. The stiffness of the coupling spring, k_3, is set to place the null of the admittance response at the 300 RPM operating state of the washing machine. *The admittance at 300 RPM is approximately* 1.2×10^{-6} $\frac{\mathrm{m}}{\mathrm{N}}$[4]

Part 3d - Displacement

Figure ?? shows the displacement of the washing machine as a function of RPM. The ramp up to 300 RPM occurs quickly so that the large displacements are transient and the maximum speed is 350 RPM (the large displacement associated with the resonance peak at higher frequencies is not reached).

The displacement at 300 RPM is approximately 38.2×10^{-6} m.

Part 3e

Figure ?? shows the force applied to the floor by the washing machine.

The force applied to the floor at 300 RPM is approximately 3.75×10^{-3} N.

[4]This is theoretical and not realistic in practical terms, but the admittance is reduced.

Problem 4 - Washing Machine Abuse Testing

The Matlab code for this problem is listed in Appendix 4.

Part 4a

1 Appendix - Matlab Code for Problem 1

2 Appendix - Matlab Code for Problem 2

3 Appendix - Matlab Code for Problem 3

4 Appendix - Matlab Code for Problem 4